RESEARCH PAPER ON ORAL CANCER DETECTION SYSTEM USING MACHINE LEARNING

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Abstract
The global incidence of oral cancer is rising at a significant rate. Early diagnosis remains the most critical factor for improving cancer survival outcomes. We introduced a high-performance machine for full-slide cytology imaging of oral cancer to support large-scale screening. The system uses a method to find the exact recovery of nuclei and then select and classify each cell using a convolutional neural network (CNN). A key innovation of the pipeline is the focus option, which enables fast and accurate decision making of each cell compared to human performance. Our results show that this method can provide effective and accurate tumor classification for full-slide cytology imaging, surpassing previous methods in terms of efficiency and effectiveness.

Keywords: CNN, Whole-slide imaging, Enormous information, Cytology, Location, Center determination, Classification.

Oropharyngeal cancers rank among the most prevalent malignancies worldwide [1, 2].Early detection of oral cancer through the examination of samples collected via brushing has been shown to effectively reduce mortality rates and support timely diagnosis.To facilitate this, we aim to integrate screening for high-risk individuals into General Dental Practices, empowering dentists and dental hygienists to conduct initial assessments. Given the extensive data involved and associated costs, computer-assisted cytological analysis is crucial for the feasibility of such an approach [4].

Whole-slide imaging (WSI) involves converting traditional microscopy slides into high-resolution digital formats through scanning. This technology is becoming increasingly popular among pathologists for its ability to improve diagnostic precision, streamline processes, and enable the integration of image data into healthcare systems [5]. However, WSI generates massive datasets—often producing images with extremely high resolution and containing large quantities of cellular structures—which creates significant challenges in terms of data management and analysis, requiring advanced techniques. Despite these challenges, WSI's ability to replicate the capabilities of conventional microscopy in a digital environment makes it a powerful tool..

Deep learning (DL) has emerged as a powerful tool for cancer classification, offering a key advantage over traditional model-based methods by eliminating the need for nucleus segmentation, a complex prerequisite for feature extraction. The extensive data generated by WSI further positions DL as a natural and effective solution. This paper introduces a fully automated, segmentation-free DL-based pipeline designed specifically for oral cancer screening using WSI.

II. Literature Review

Several studies advocate the use of deep learning (DL) for classifying histology whole-slide imaging (WSI) samples [6–9]. A commonly used technique is to break down large whole-slide images into smaller sections for detailed examination. However, there are significant differences between cell analysis in tissue versus in other types of preparations. In tissue examination, the positioning and relationships between cells within the specimen are critical, whereas other forms of analysis may focus on different factors making region segmentation and processing logical. Conversely, cytological samples, particularly liquid-based ones, consist of individual cells with no extracellular structure, making the cell the natural unit of analysis.

Cytology also demands higher resolution than histology, as texture details are crucial, and precise focus is necessary. Slide scanner auto-focus works more effectively with tissue samples due to their relatively flat surfaces. In cytological samples, overlapping cells and varying z-levels create challenges, as tissue analysis tools often lack support for z-stacks (focus level stacks) or features to handle such complexities. This work introduces a carefully developed pipeline tailored for processing cytological WSIs with multiple focus levels. It includes steps for Identification of individual cells, determination of optimal focus for each cell, and categorization using convolutional neural networks (CNNs).

Malignancy-related alterations (MRAs) refer to slight morphological variations in cells that appear normal under a microscope but are located near tumor areas These changes have been consistently measurable across various cancer types through image cytometry [10], suggesting their potential tools to mark A classification tool has been shown to reliably identify MACs in Visually typical oropharyngeal epithelial cells located near tumors, offering non invasive approach for detecting early-stage oropharyngeal cancers. Leveraging MACs allows for patient-level diagnosis to train cell-level classifiers, assigning all cells within a patient a single label (cancerous or healthy) [12]. This significantly alleviates the otherwise time-intensive and challenging task of manual cell-level annotation.

Research Methodology & Experiment Design

A. Problem Definition

objective is to substantiate a Oral Cancer detection tool having following features

- Cell detection:
- Focus Selection: In cytological analysis, it is necessary to determine the focus level for each nucleus individually, as cells are positioned at varying depths.
- Classification: In cytology, classification involves identifying and categorizing individual cells based on their morphological features.

3.1. Data

This research makes use of three distinct datasets consisting of images obtained from oral cell collection samples. The first dataset includes a small collection of Pap smear images captured with a conventional microscope. The second dataset contains whole-slide images (WSIs) produced from the same glass slides used in Dataset 1, while the third dataset consists of WSIs created from slides prepared using liquid-based cytology (LBC). All data were gathered at a study conducted in Stockholm, oral samples were collected from participants using a brush to target specific areas inside the mouth. These samples were either directly transferred to glass slides (Datasets 1 and 2) or placed in liquid containers for further analysis (Dataset 3). The samples underwent staining following the traditional Papanicolaou technique, while the slides in Dataset 3 were processed using a commercial ThinPrep system, following standard procedures for non-gynecological cytology.

For Dataset 1, images were captured using a microscope system equipped with a 20× magnification objective, resulting in a pixel resolution of 0.32 µm. This dataset contains isolated cell images from 10 different Pap smear slides, each from a unique participant. The images were cropped into grayscale sections of 80×80×1 pixels, with the nucleus precisely focused in the center of each patch.

Dataset 2 consists of color (RGB) images of the same slides from Dataset 1. These were obtained using a slide scanning system with a 40× objective lens and a numerical aperture of 0.75. The scanning process involved 11 z-axis offset levels, each separated by 0.4 µm. The resulting images have dimensions of 103,936×107,520 pixels and a pixel resolution of 0.23 µm.

Dataset 3 was gathered similarly to Dataset 2 but includes an additional 12 liquid-based cytology slides from 12 new participants.

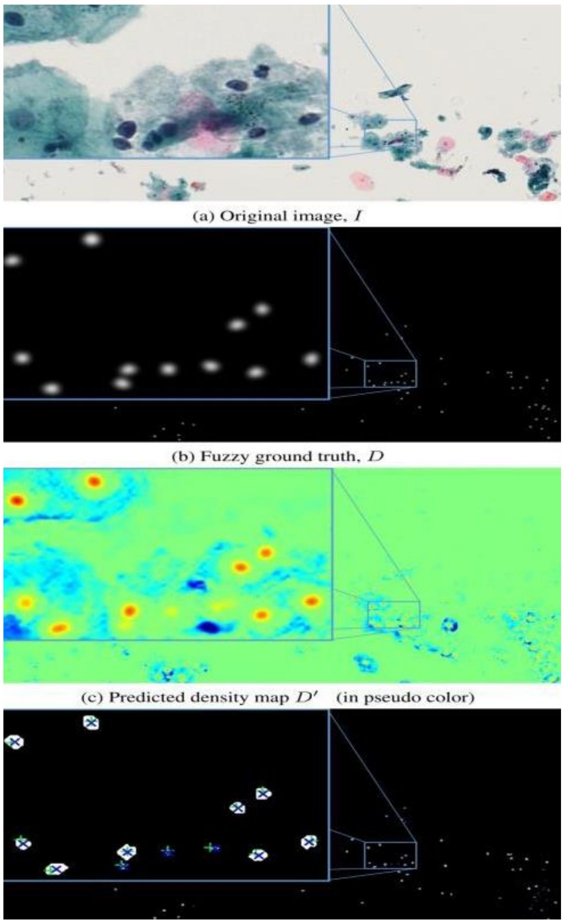
To reduce the significant effort required for manual annotations at the cellular level, annotating entire slides and focusing on malignancy-associated changes (MACs) presents a more feasible solution. Previous research [11, 12] has demonstrated encouraging outcomes in detecting MACs across both tissue and cell-based samples. Building on these findings, our objective is to categorize cells according to patient diagnoses, designating all cells from individuals with oral cancer as malignant.

Nucleus Localization

The process of detecting nuclei centers on precisely locating the individual cell nuclei within whole-slide images (WSIs), adapting concepts from the Fully Convolutional Regression Networks (FCRNs) technique used for cell quantification [22]. The critical stages of this method are summarized below, with an example shown from Dataset 3 (Fig. 1)

Model Learning Process:

The input consists of a series of RGB images paired with their corresponding binary labels, where each nucleus is represented by a central pixel. To generate a version of the label, each label is expanded using a circular shape with a specified size, followed by smoothing through a filter with a defined parameter. A deep model is then trained to map the original image (Fig. 1a) to this version of the label (Fig. 1b). The model is built on a segmentation framework [30], with the last layer using a different activation function.



(a) Original image, I

(b) Fuzzy ground truth, D

(c) Predicted density map D' (in pseudo color)

Prediction Process:

For each input image, a density map (Fig. 1c) is produced as the model's output. This map is then subjected to a thresholding operation at a predetermined value, and the centroids of the resulting regions are used to identify the locations of the nuclei.

Focus Determination

For cell analysis, a specialized focus determination procedure is necessary, as slide scanners alone often do not provide sufficient clarity. The proposed approach utilizes multiple z-planes captured at equal intervals for the same sample. By examining the differences between successive images at these levels, the point with the highest variance indicates the transition from sharp to blurred focus. This novel approach to focus determination offers a substantial advancement compared to the previously used **edge-based blur assessment method**

Once the nuclei are located at the most optimal focal plane, a rectangular area surrounding each detected nucleus is cropped from all available z-planes. Each cropped section is then processed with a noise-reduction technique using a small median filter applied to each individual color layer. This results in a set of images that represent the same nucleus across the different focal planes

$$\sigma_i^2 = \frac{1}{M}\sum_{j=1}^{M}\left(p'_{ij}-\mu_i\right)^2 \text{, where } \mu_i = \frac{1}{M}\sum_{j=1}^{M} p'_{ij}$$

Categorization

The last step in the process focuses on classifying the extracted cell regions into two categories: malignant / non-malignant.. In line with the approach outlined in [12], we evaluate the ResNet50 model [31], as well as the more recent DenseNet201 architecture [32]. Both models are tested using weights initialized with the Glorot-uniform method, alongside pre-trained weights from ImageNet for comparison.

Considering the crucial role of texture features in distinguishing between categories [11, 33], the dataset is augmented without using interpolation techniques. During the training process, each image has a 50% chance of being mirrored, and is randomly rotated by multiples of 90° to increase diversity.

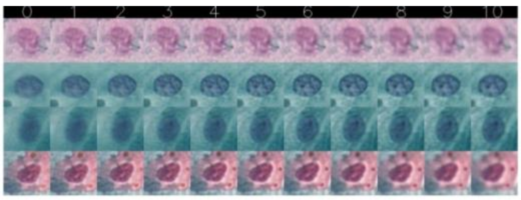


Fig. 2: Example of focus sequences for experts to annotate

Experimental Configuration

Nucleus Localization

For nucleus detection, images from the central focal plane are selected. Each whole-slide image (WSI) is divided into smaller sections of size 5120×2560×3 using a public tool [34]. The model is trained on 15 sections containing a total of 850 nuclei and evaluated on 3 sections with 150 nuclei from Dataset 3. To prepare the reference data, labeled annotations are expanded using a circular shape with a radius of r. All images, along with the reference annotations, are resized to 1024×512 pixels using a technique that preserves area. A smoothing filter with a specified standard deviation is then applied to the reference masks to create a blurred version

Before training, images are standardized using the mean and scaled. Various augmentation techniques are applied, including rotations. (ranging from ±30°), random shifts along the horizontal and vertical axes (up to 30% of the image dimensions), random zooming (up to a 30% change), and random horizontal and vertical flipping. Since texture details are not critical for detecting nuclei, interpolation does not influence the model's performance. To stabilize the learning process, normalization is applied before each activation function [35]

The model is optimized using the RMSprop algorithm, aiming to reduce the difference between predicted and actual values. The training procedure is carried out over 100 iterations with a batch size of 1, and the model that exhibits the smallest error is selected for testing. The performance is assessed using the test dataset, where a match is considered accurate when the identified cell falls within the area of the nearest reference label. Each reference cell is linked to at most one detected object

4.2. Focus Determination

For focus evaluation, 100 cells were selected. Each cell was extracted as an 80×80×3 patch from all 11 focal planes. In the case of the EMBM approach, the threshold for detecting a clear edge was set according to the method in [27]. To identify the best focus level, 8 specialists were asked to review and select the most appropriate focus for each of the 100 cells from the 11 focal planes (Fig. 2). The focus level chosen by the majority of experts was taken as the reference or ground truth. A predicted focus level was considered acceptable if it fell within a specified range around the ground truth focus level.

Categorization

The performance of the categorization framework was first evaluated using Dataset 1 as a reference point. It was then applied to Dataset 2 to assess how effectively the components for cell detection and focus identification performed, compared to the results from Dataset 1. Finally, the system was tested on Dataset 3 to verify its adaptability and robustness across different datasets. For all datasets, patient-level splits were used to ensure no overlap between the data used for training and testing. Three-fold cross-validation was performed for Datasets 1 and 2, in line with the method outlined in [12], while two-fold cross-validation was applied to Dataset 3.

For Datasets 2 and 3, nucleus patches were generated using the trained nucleus detector with a threshold of , identified as optimal in Section 5.1. Cells outside the ±2 μm imaged z-level range in Datasets 2 and 3 tended to be blurred even at the best focus level. To exclude overly blurred cells, the EMBM method was applied, and patches with an EMBM score below 0.03 were discarded. This filtering left 68,509 cells in Dataset 2 and 130,521 in Dataset 3.

The models were optimized using a gradient-based optimization method, minimizing the loss function that measures the difference between predicted and true class labels , following parameters outlined in [36]: an initial learning rate , , and . A validation set comprising 10% of the training data was randomly selected.

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For models initialized with pre-trained weights from ImageNet, the single-channel images from Dataset 1 were replicated across three color channels to meet the model's input requirements. These models underwent a fine-tuning process for five iterations, with the learning rate reduced by 40% if the validation error did not improve compared to the previous iteration. The least error condition was chosen for testing.

A different procedure was followed for models trained from scratch. The ResNet50 architecture was trained for 50 iterations with a mini-batch size of 512 on Datasets 2 and 3, and for 30 iterations with a batch size of 128 on Dataset 1 due to its smaller size. The DenseNet201 model, which demands more GPU resources, was trained with a batch size of 256 on Datasets 2 and 3, and fewer iterations were performed to avoid overfitting (30 iterations on Datasets 2 and 3, and 20 iterations on Dataset 1). For DenseNet201, growth reduced 90% .In case zero progress happened in validation loss over five iterations, and training was halted if zero growth was observed in 15 iterations. The system with least disadvantage was retained.

'able 1: Classification performance. The best F1-score for ach dataset is presented in bold.

Dataset	Network	Accuracy	Precision	Recall	F1-score
1	ResNet50	70.5±0.5	63.1±1.2	34.8±1.4	44.8±1.3
	ResNet50(pre-trained)	72.0±0.9	66.4±2.0	37.5±2.0	**48.0±2.1**
	DenseNet201	70.4±0.5	63.1±1.8	33.8±0.9	44.0±0.7
	DenseNet201(pre-trained)	70.6±0.7	63.4±1.6	34.3±1.7	44.5±1.8
2	ResNet50	74.4±1.9	83.3±2.9	46.3±3.8	59.5±3.8
	ResNet50(pre-trained)	74.0±0.1	83.9±0.5	44.6±0.7	58.2±0.5
	DenseNet201	75.4±0.8	84.3±1.5	48.3±1.1	**61.4±1.3**
	DenseNet201(pre-trained)	73.3±0.7	81.7±2.8	44.4±0.3	57.5±0.6
3	ResNet50	81.6±0.7	71.7±1.2	73.8±0.9	**72.8±1.0**
	ResNet50(pre-trained)	81.3±1.5	72.1±3.0	71.6±0.6	71.8±1.8
	DenseNet201	81.3±0.5	71.4±0.7	73.0±0.8	72.2±0.7
	DenseNet201(pre-trained)	81.5±1.3	71.2±2.4	74.5±2.4	72.8±1.9

OUTCOMES AND ANALYSIS

Figure 3 displays the results of the cell detection process. The graph shows how key performance metrics change when the threshold is adjusted within a specified range. At a threshold value of 0.59, the model achieves the highest performance, with detection values being significantly strong. At this level, 94,685 individual cells were found in **Dataset 2**, while 138,196 were identified in **Dataset 3**. The time taken to create a density map is minimal, requiring just 0.17 seconds using a specific GPU. In contrast, an alternate method for generating a density map takes much longer to process.
The effectiveness of the focus selection method is also shown in **Figure 4**. Here, expert results, which were averaged from several specialists, highlight the performance of a blur metric technique when using the closest focus levels. As certain parameters are increased, the method's performance approaches that of the alternative approach. While the blur metric alone does not provide optimal results, enhancing it with a median filter produces significant improvement. The accuracy of our approach is very close to that of the human experts.
The results from the classification task are presented in **Table 1** and **Figure 5**. Both the ResNet50 and DenseNet201 architectures yielded similar results. Pre-training provided a small advantage with the smaller dataset, but had no notable effect on the larger datasets. The outcomes for **Dataset 2** were notably better than for **Dataset 1**, highlighting the success of the cell detection and focus selection processes. By including a larger set of nuclei, the performance was also significantly improved. **Dataset 3** results confirmed the model's ability to generalize well to data from liquid-based cytology methods.
In **Figure 6**, we show how accuracy declines when cells are intentionally selected from non-optimal focus levels. The drop in performance as the focus deviates from the optimal setting underscores the critical role of precise focus selection.
When combining the classifications of cells across entire slides, as seen in **Figure 5**, it's clear that slides which were initially indistinguishable in **Dataset 1** become clearly separable in **Dataset 2**. By setting global thresholds, we can effectively differentiate the two patient groups across the datasets processed through our system.

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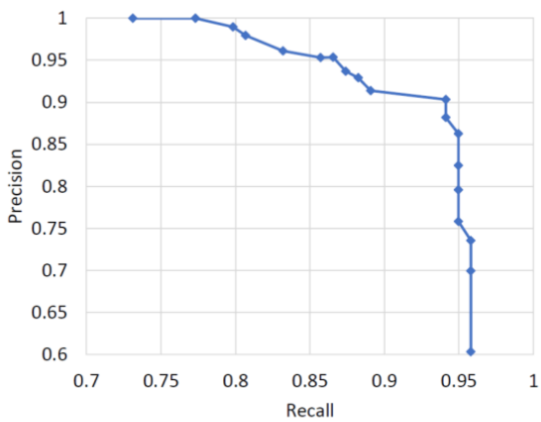

(b) Precision-recall curve

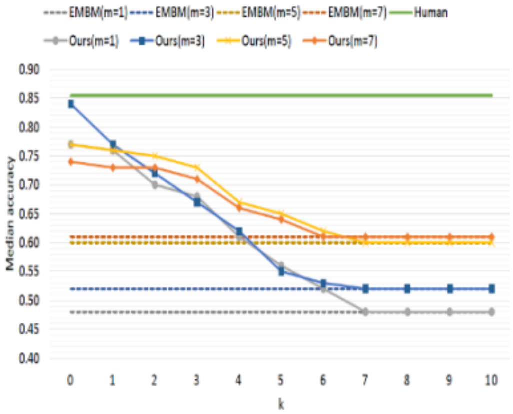

Fig. 4: Accuracy of focus selection

Conclusion:

To summarize, this study introduces a fully automated framework designed for detecting oral cancer using whole slide images. The implementation, built with TensorFlow 1.14, is made publicly accessible for research and development purposes. The technique employed for selecting the optimal focus demonstrates performance comparable to human experts, providing results that surpass the traditional blur-based approach. The system efficiently performs analysis on whole slide images within a reasonable time, exhibiting consistent results across both smear and liquid-based cytology samples.

By comparing the results from **Dataset 1**, where nuclei are manually chosen, to **Dataset 2**, where automated methods are used to identify nuclei from the same slides, the proposed pipeline proves to reduce the need for manual labor, while enhancing both the precision and consistency of the classification process.

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    - This paper, led by Speight and others, presents key insights from the Global Oral Cancer Forum, addressing the need for more effective screening methods and offering recommendations based on expert consensus.